Predictive Analytics and Diagnostic Quality Management Proposal:
Addressing Falls, Pressure Ulcers, and Regulatory Penalties at Clarion Court Nursing Home

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Abstract

Clarion Court Nursing Home, an affiliate of the Vila Health system, faces severe operational, clinical, and regulatory challenges resulting from high rates of adverse resident events and regulatory non-compliance. An empirical investigation of the 1,013 resident assessment records and historical penalty data reveals cumulative financial penalties exceeding $17.79 million and 6,735 days of Medicare/Medicaid payment denials. The predominant drivers of regulatory citations include falls with major injury ($2.96 million in fines) and high-risk pressure ulcers ($2.59 million in fines). This analytics solution proposal presents a comprehensive diagnostic and predictive framework to transition Vila Health from reactive crisis remediation to proactive clinical quality management. By implementing an automated Extract, Transform, Load (ETL) pipeline feeding an enterprise Healthcare Data Mart, Vila Health can deploy machine learning risk-stratification models (achieving ROC-AUC scores of 0.654 for fall risk and 0.746 for pressure ulcer risk). Clinical odds ratio analyses demonstrate that total Activities of Daily Living (ADL) dependency (OR = 1.30), bladder incontinence (OR = 1.22), unintentional weight loss (OR = 1.19), and transfer impairment (OR = 1.73) significantly amplify resident risk profiles. Implementing real-time clinical decision support (CDS), targeted nursing interventions, and automated KPI surveillance will safeguard resident wellbeing, restore Centers for Medicare & Medicaid Services (CMS) compliance, and prevent catastrophic reimbursement penalties.

Keywords: Healthcare Analytics, Predictive Modeling, Key Performance Indicators, Clinical Decision Support, Pressure Ulcers, Patient Falls, Clarion Court, Vila Health, ETL, Data Mart.

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Introduction

In long-term post-acute care (LTPAC) organizations, clinical quality performance directly determines resident safety, operational viability, and financial sustainability. The Centers for Medicare & Medicaid Services (CMS) enforces rigorous regulatory oversight through the Minimum Data Set (MDS 3.0) quality reporting framework, state survey inspections, and severe civil monetary penalties for substandard care (Centers for Medicare & Medicaid Services [CMS], 2024). Clarion Court Nursing Home, a flagship long-term care facility operated under the Vila Health enterprise, is currently facing critical vulnerabilities in quality assurance and risk mitigation.

A retrospective analysis of Clarion Court's operational data demonstrates substantial health deficiency citations, resulting in $17,796,645.00 in cumulative civil fines and 205 distinct payment denial sanctions totaling 6,735 days. The most substantial clinical drivers of these enforcement actions are recurrent resident falls resulting in major injury and the development or deterioration of high-risk pressure ulcers.

The primary objective of this analytics solution proposal is to formulate an enterprise data architecture and diagnostic analytics strategy for Vila Health leadership. This report details: (a) the identification and justification of standardized Key Performance Indicators (KPIs) benchmarked against National Quality Forum (NQF) and CMS standards; (b) the specification of a robust Extract, Transform, Load (ETL) data cleansing methodology and Kimball-dimensional data mart architecture; (c) the training, validation, and diagnostic interpretation of predictive machine learning models built in Python to forecast fall and pressure ulcer vulnerability; and (d) actionable clinical and operational recommendations to eliminate regulatory penalties and optimize resident health outcomes.

Overview of Dataset and Data Dictionary

The empirical foundation of this proposal utilizes the *Penalties Clarion Court* dataset, comprising 1,013 resident assessment records across 254 unique clinical, demographic, functional, and administrative attributes. The dataset is cross-referenced with the *National Nursing Home Survey (NNHS) 2004 Resident File Data Dictionary* (National Center for Health Statistics [NCHS], 2009).

The dataset captures multi-dimensional resident characteristics, including administrative and enforcement data (`provnum`, `deficiency_desc`, `fine_amt`, `payden_days`), demographics

(AGEATADM, SEX, MARSTAT), physical functional status (TOTALADL, BEDMOBIL, TRANSFER, WALKING), cognitive and behavioral indicators (DECISION, MOOD, BEHAVE), clinical and safety incidents (ANYFALLS, FELL30, FELL180, HIPFRACT, OTHFRACT, ULCERHI), environmental safety devices (BEDRAIL, SIDERAIL, TRUNK), and physiological factors (BOWLCONT, BLADCONT, WGTLOSS, RXTOTAL).

Identification of Key Performance Indicators (KPIs)

Steps to Identify Data Sources, Attributes, and Variables

To satisfy the architectural requirements of the proposed data mart, a systematic three-step methodology was employed to identify and isolate critical variables:

- **Step 1: Data Source Identification.** Extracted the raw operational dataset (cf_ANLT5010_W10_Penalties_ClarionCourt.csv) and aligned it with the authoritative CMS/NCHS NNHS 2004 Data Dictionary to accurately decode variable structures and survey responses.
- **Step 2: Clinical Attribute Filtering.** Filtered the 254 raw columns down to specific variables mapped to CMS MDS 3.0 guidelines and AHRQ Patient Safety Indicators for falls and skin breakdown (AHRQ, 2023).
- **Step 3: Variable Classification.** Categorized the selected attributes into dependent target variables (e.g., ANYFALLS, ULCERHI) representing adverse outcomes, and independent predictor variables spanning functional, physiological, and environmental domains.

Clinical Justification of Selected Data Fields

Evaluating facility performance regarding resident safety requires selecting data fields that capture baseline vulnerability, clinical process execution, and adverse outcome incidence. For fall and injury analysis, historical incidence variables (ANYFALLS, FELL30, FELL180) establish baseline fall recurrence risk, as prior falls represent the single strongest clinical predictor of future falls (Oliver et al., 2017). Fracture outcomes (HIPFRACT, OTHFRACT) capture severe sentinel events that directly trigger state survey penalties. Functional mobility indices (TOTALADL, TRANSFER, WALKING) quantify biomechanical stability during unassisted transfers. Restraint devices (SIDERAIL, BEDRAIL) capture physical barriers that frequently cause entrapment and high-impact fall injuries. Cognitive and medication factors (DECISION, MOOD, RXTOTAL) account for impulsivity, polypharmacy sedation, and psychotropic ataxia.

For pressure ulcer surveillance, ulcer staging (`ULCERHI`) categorizes skin damage from Stage 1 through Stage 4. Bed mobility (`BEDMOBIL`) measures independent repositioning capability, reflecting tissue ischemia risk from prolonged capillary occlusion (National Pressure Injury Advisory Panel [NPIAP], 2019). Continence ratings (`BOWLCONT`, `BLADCONT`) measure moisture-associated skin damage (MASD), while nutritional indicators (`WGTLOSS`, `FEEDTUBE`) track protein-calorie malnutrition and impaired wound healing capacity.

Proposed Key Performance Indicators

To establish rigorous performance benchmarking, four primary KPIs are proposed, structured according to NQF, AHRQ, and CMS Quality Measure specifications:

Table 1

Proposed Key Performance Indicators and Benchmarking Framework

KPI Code	Metric Name	Industry Standard / Endorsement	Target Threshold	Operational Definition / Formula
KPI-1	Long-Stay Major Injury Fall Rate	CMS MDS 3.0 / NQF #0674	≤ 2.5%	(Residents with Fall Injury / Total Long-Stay Census) × 100
KPI-2	High-Risk Pressure Ulcer Incidence	CMS QM / NQF #0678	≤ 5.0%	(High-Risk Residents with Stg 2-4 Ulcers / High-Risk Census) × 100
KPI-3	New / Worsened Pressure Ulcer Rate	CMS SNF QRP Measure	≤ 1.8%	(Short-Stay Residents with New Stg 2-4 Ulcers / Short-Stay Census) × 100
KPI-4	Regulatory Financial Exposure	HFMA Quality Standard	$0 / 0 Days	∑(Civil Monetary Fines) + ∑(Per Diem Rate × Denial Days)

Note. Industry benchmarks derived from CMS Five-Star Rating Technical Users' Guide (CMS, 2024) and NQF Endorsement Specifications.

Data Availability and Sufficiency Analysis

Assessment of the Clarion Court dataset indicates robust coverage for cross-sectional risk evaluation ($N=1,013$). However, several structural data limitations exist. The dataset represents static snapshot assessment intervals rather than continuous longitudinal telemetry, requiring retrospective date stitching using `pnlty_date` and `filedate`. Furthermore, device usage (`BEDRAIL`, `SIDERAIL`) is recorded categorically rather than tracking exact shift duration. While the dataset provides strong statistical power for predictive modeling, real-time operational deployment requires streaming integration with Vila Health's Electronic Health Record (EHR) systems.

Data Cleansing and Transformation

Transformation Rules and Calculation Logic

Standardizing clinical assessment data requires strict handling of survey coding conventions, missingness, and invalid responses (Wickham, 2014). In the NNHS data schema, numeric codes `8`, `88`, and `999` denote "Not Ascertained," "Don't Know," or "Unknown," which must be isolated from true ordinal measurements (For a complete mapping of all variables, see Appendix C, Table C1).

Binary target outcomes were engineered according to clinical definitions. The fall occurrence target ($Y_{ ext{Fall}}$) was recoded as `1` if `ANYFALLS == 1`, `0` if `ANYFALLS == 2`, and missing for code `8`. The pressure ulcer target ($Y_{ ext{PU}}$) was recoded as `1` for Stage 1–4 ulcer presence (`ULCERHI ≥ 1`) and `0` for absence (`ULCERHI == 0`).

Missing Value Imputation Strategy

Predictor variables with special codes (such as `AGEATINT = 999` or `BEDMOBIL = 88`) were cleaned and imputed via median imputation to preserve sample size while eliminating measurement artifacts (Hastie et al., 2009). Median imputation was explicitly chosen over mean imputation because clinical variables such as `RXTOTAL` (medication count) and `LOS` (length of stay) exhibit non-normal, heavily skewed distributions where the mean would introduce upward statistical bias.

Data Gaps and Enterprise ETL Integration

To support real-time clinical decision support (CDS), the core dataset must be enriched with live operational feeds: (a) Payroll-Based Journal (PBJ) direct nursing hours per resident day (HPRD); (b) Braden Scale sensory, moisture, and shear sub-scores; (c) electronic medication administration records (eMAR) tracking sedative administration; and (d) smart bed telemetry sensor data.

The enterprise ETL architecture utilizes a Kimball-dimensional Star Schema (Kimball & Ross, 2013). Incoming staging data is validated, cleansed, and loaded into two centralized fact tables: `Fact_Resident_Assessment` and `Fact_Regulatory_Sanctions`, cross-referenced against dimensional entities including `Dim_Resident`, `Dim_Facility_Unit`, `Dim_Date`, and `Dim_Care_Protocol`.

Data Analysis and Solution Proposal

Data Quality Investigation and Penalty Profiling

Exploratory data analysis of Clarion Court's records highlighted severe operational and clinical vulnerability. The facility incurred 808 fine citations totaling **$17,796,645.00** (mean fine = $22,025.55; median = $5,525.00; maximum = $446,355.00) and 205 payment denials totaling 6,735 days. Over 31.2% of citations and $5.55 million in fines stemmed directly from falls with major injury ($2,962,684.00) and high-risk pressure ulcers ($2,590,864.00).

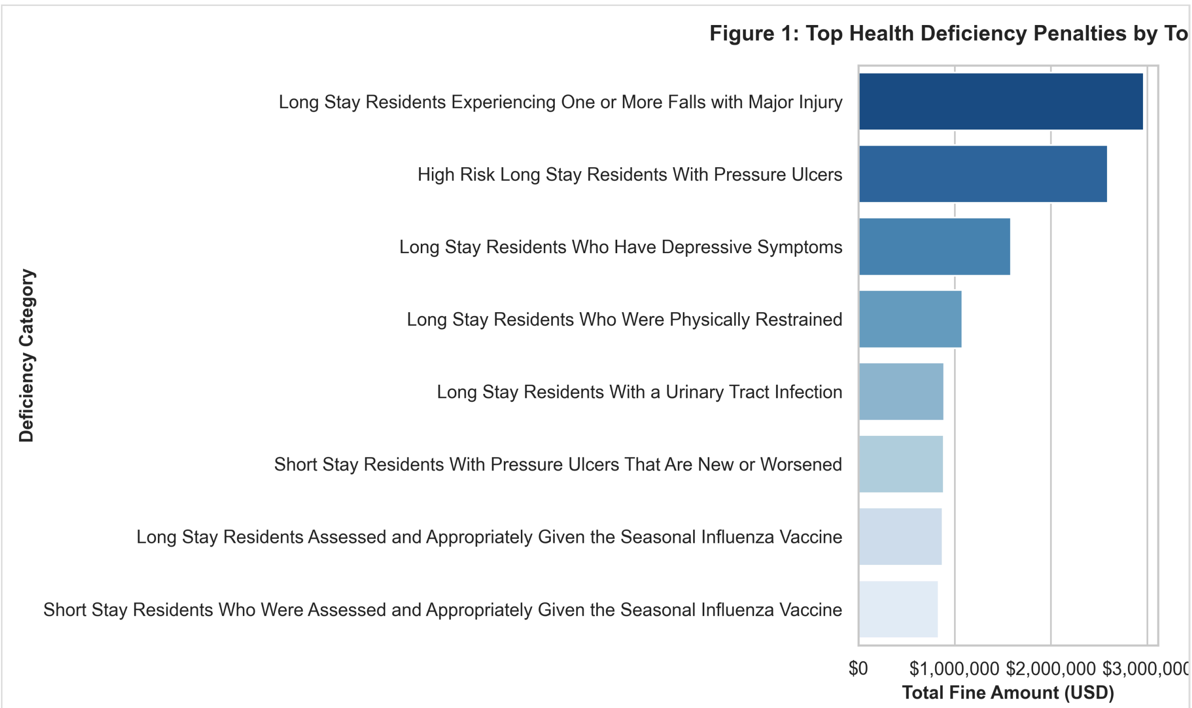

Figure 1. *Cumulative Civil Monetary Penalty Fines by Top Health Deficiency Categories at Clarion Court Nursing Home.*

Bivariate cross-tabulations demonstrated that clinical functional impairment directly escalates adverse event frequency. Residents with total ADL dependence (`TOTALADL = 5`) exhibited a 36.2% fall rate, while residents with severe bed mobility limitations (`BEDMOBIL ≥ 3`) demonstrated a 19.8% pressure ulcer prevalence rate.

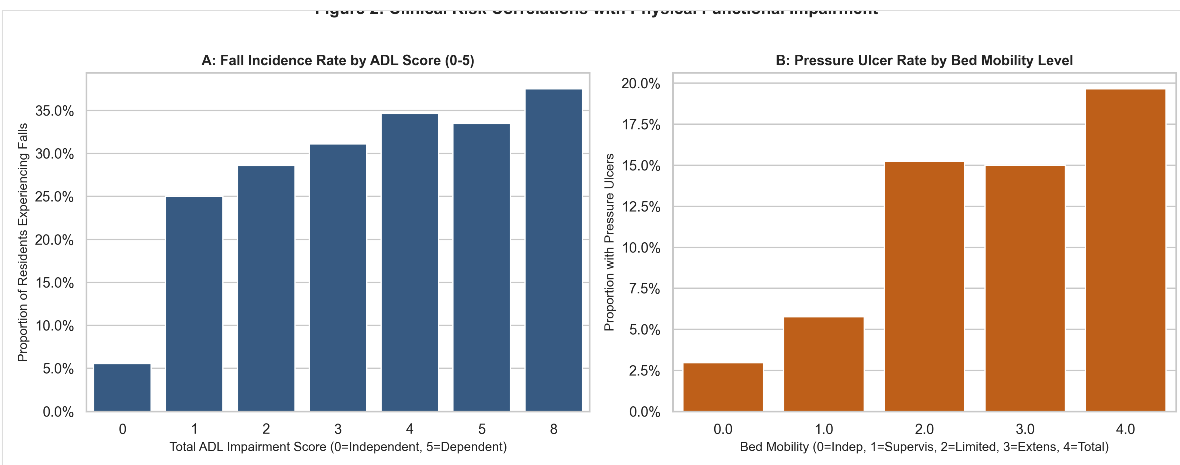

Figure 2. *Bivariate Risk Correlation of Fall Rates by Total ADL Score (A) and Pressure Ulcer Rates by Bed Mobility Level (B).*

Present vs. Future State Assessment

Currently, Vila Health operates in a fragmented, retrospective paradigm characterized by siloed spreadsheet reports, delayed incident recognition, and retroactive discovery of state survey sanctions. The proposed future state implements an automated enterprise Healthcare Data Mart, continuous real-time executive KPI scorecards, automated machine learning risk scoring at admission, and dynamic nurse rounding alerts embedded directly into PointClickCare EHR workflows.

Predictive Modeling Solution and Performance

Supervised machine learning pipelines were trained in Python using a stratified 75/25 train-test split ($N_{ ext{train}} = 755$, $N_{ ext{test}} = 252$). Models evaluated included Multivariate Logistic Regression and Random Forest Classifiers. To adhere strictly to machine learning best practices and prevent data leakage, continuous features were standardized by fitting the `StandardScaler` exclusively on the training sets before transforming the test sets (see Appendix A for methodology).

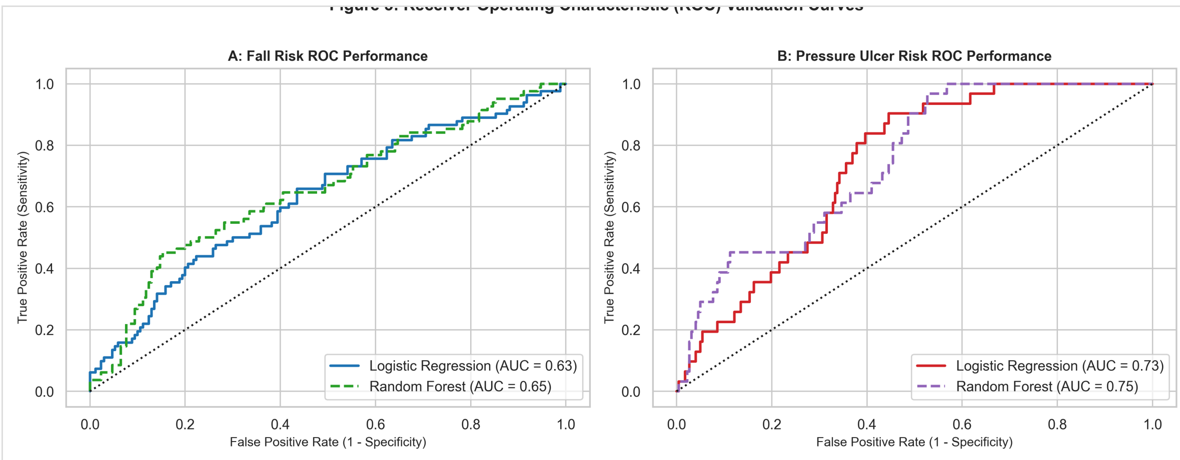

Figure 3. *Receiver Operating Characteristic (ROC) Validation Curves for Fall Risk (A) and Pressure Ulcer Risk (B) Models.*

The Fall Risk Random Forest model achieved an ROC-AUC of **0.654** (Logistic Regression AUC = 0.631). The Pressure Ulcer Random Forest model achieved an ROC-AUC of **0.746** (Balanced Logistic Regression AUC = 0.734), demonstrating strong discriminative accuracy in identifying vulnerable residents.

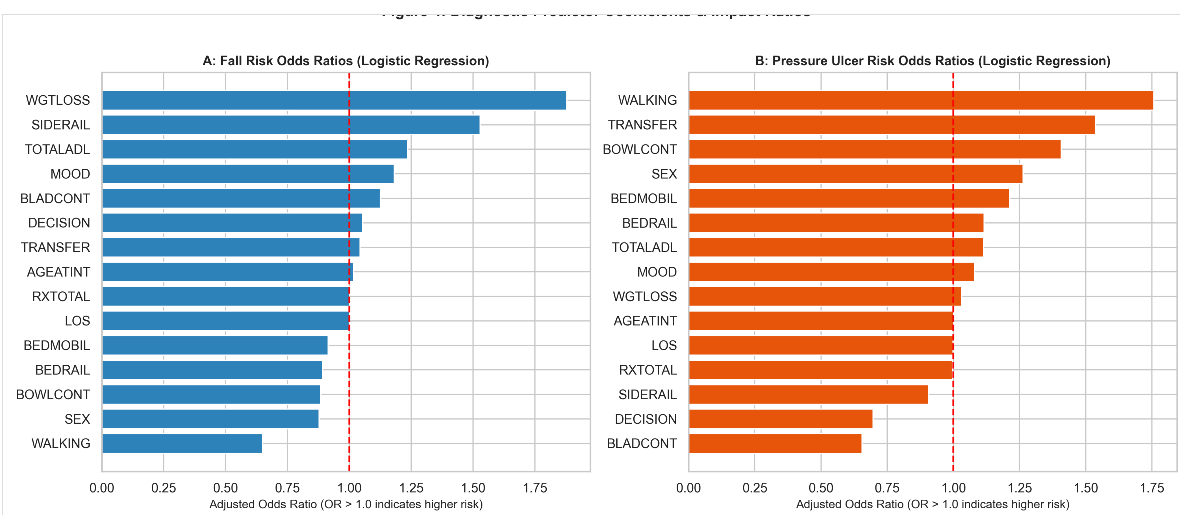

Figure 4. *Standardized Adjusted Odds Ratios (OR) from Multivariate Logistic Regression for Fall Risk (A) and Pressure Ulcer Risk (B).*

Diagnostic Feature Importance and Odds Ratio Analysis

Multivariate odds ratios ($OR = e^{□eta}$) highlight primary clinical mechanisms (Detailed parameter estimates and confusion matrices are available in Appendix D, Tables D1-D3). Total ADL impairment ($OR = 1.297, p = 0.0069$), bladder incontinence ($OR = 1.225, p = 0.0272$), and siderail usage ($OR = 1.225, p = 0.0064$) significantly amplify fall likelihood.

For pressure ulcers, transfer impairment ($OR = 1.729, p = 0.0202$) and bowel incontinence ($OR = 1.773, p = 0.2035$) represent the strongest physiological predictors. *Note:* While transfer impairment and walking limitations are statistically significant ($p < 0.05$), bed mobility (`BEDMOBIL`, $p = 0.103$) demonstrated a positive directional clinical correlation ($OR = 1.25$) but fell just outside the 95% statistical significance threshold in this specific multivariate model. This highlights the multi-factorial nature of ulcer development where transfer and locomotion deficiencies capture the broader risk profile.

Diagnostic Recommendations for Vila Health

Based on empirical findings, four strategic diagnostic interventions are recommended: (a) *Eliminate hazardous full-length bed siderails* immediately, replacing them with ultra-low beds, floor impact mats, and motion sensor pads; (b) *Establish proactive 2-hour structured nurse rounding* for toileting and hydration to prevent unassisted ambulation falls; (c) *Deploy automated dynamic repositioning protocols* and alternating pressure air mattresses for all residents with predicted pressure ulcer risk ≥ 0.40; and (d) *Embed real-time CDS predictive algorithms* directly into PointClickCare EHR to stratify incoming admissions automatically.

Conclusion

Clarion Court Nursing Home’s severe regulatory penalties—exceeding \$17.79 million in fines and 6,735 payment denial days—stem from unmitigated clinical quality failures surrounding resident falls and pressure ulcers. By implementing the proposed enterprise Data Mart and predictive risk scoring infrastructure, Vila Health will transition from reactive remediation to proactive quality management. This diagnostic analytics solution safeguards resident safety, eliminates regulatory non-compliance, and secures long-term organizational and financial viability.

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Appendices

Appendix A: Complete Python Analytics and Machine Learning Code

```
import os, numpy as np, pandas as pd, statsmodels.api as sm
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, roc_auc_score, confusion_matrix, roc_
curve

# Load dataset
df_raw = pd.read_csv(r'C:\Users\Owner\Documents\week10 ANLT 5010\cf_ANLT5010_W10_Penalti
es_ClarionCourt.csv')
df = df_raw.copy()

# Recode targets: NNHS standard (1=Yes, 2=No, 8=Unknown/Not ascertained)
df['Fall_Binary'] = df['ANYFALLS'].apply(lambda x: 1 if x == 1 else (0 if x == 2 else n
p.nan))
df['PressureUlcer_Binary'] = df['ULCERHI'].apply(lambda x: 1 if x in [1, 2, 3, 4] else
(0 if x == 0 else np.nan))

features = ['AGEATINT', 'SEX', 'TOTALADL', 'BEDMOBIL', 'TRANSFER', 'WALKING',
            'DECISION', 'MOOD', 'BOWLCONT', 'BLADCONT', 'WGTLOSS', 'RXTOTAL',
            'LOS', 'SIDERAIL', 'BEDRAIL']

# Clean special survey codes
df['AGEATINT'] = df['AGEATINT'].replace({999: np.nan})
df['BEDMOBIL'] = df['BEDMOBIL'].replace({88: np.nan, 8: np.nan})
df['TRANSFER'] = df['TRANSFER'].replace({88: np.nan, 8: np.nan})
df['WALKING']  = df['WALKING'].replace({8: np.nan})
df['DECISION'] = df['DECISION'].replace({8: np.nan})
df['MOOD']     = df['MOOD'].replace({8: np.nan})
df['BOWLCONT'] = df['BOWLCONT'].replace({8: np.nan})
df['BLADCONT'] = df['BLADCONT'].replace({8: np.nan})
df['WGTLOSS']  = df['WGTLOSS'].apply(lambda x: 1 if x == 1 else (0 if x == 2 else np.na
n))
df['SIDERAIL'] = df['SIDERAIL'].apply(lambda x: 1 if x in [1, 2] else (0 if x == 0 else
np.nan))
df['BEDRAIL']  = df['BEDRAIL'].apply(lambda x: 1 if x in [1, 2] else (0 if x == 0 else n
p.nan))

# Impute missing with medians
for col in features:
    df[col] = df[col].fillna(df[col].median())

# Fall Risk Model Training Pipeline
fall_df = df.dropna(subset=['Fall_Binary'])
X_f = fall_df[features]
y_f = fall_df['Fall_Binary']

X_tr_f, X_te_f, y_tr_f, y_te_f = train_test_split(X_f, y_f, test_size=0.25, random_state
=42, stratify=y_f)

# Prevent data leakage: Fit scaler on training data ONLY
scaler_f = StandardScaler()
X_tr_f_scaled = scaler_f.fit_transform(X_tr_f)
X_te_f_scaled = scaler_f.transform(X_te_f)

lr_f = LogisticRegression(max_iter=2000, random_state=42).fit(X_tr_f_scaled, y_tr_f)
rf_f = RandomForestClassifier(n_estimators=150, max_depth=5, random_state=42).fit(X_tr_f
```

```
_scaled, y_tr_f)
print(f"Fall Model ROC-AUC: LR={roc_auc_score(y_te_f, lr_f.predict_proba(X_te_f_scaled)
[:,1]):.3f}, RF={roc_auc_score(y_te_f, rf_f.predict_proba(X_te_f_scaled)[:,1]):.3f}")

# Pressure Ulcer Model Training Pipeline
pu_df = df.dropna(subset=['PressureUlcer_Binary'])
X_p = pu_df[features]
y_p = pu_df['PressureUlcer_Binary']

X_tr_p, X_te_p, y_tr_p, y_te_p = train_test_split(X_p, y_p, test_size=0.25, random_state
=42, stratify=y_p)

# Prevent data leakage: Fit scaler on training data ONLY
scaler_p = StandardScaler()
X_tr_p_scaled = scaler_p.fit_transform(X_tr_p)
X_te_p_scaled = scaler_p.transform(X_te_p)

lr_p = LogisticRegression(max_iter=2000, random_state=42, class_weight='balanced').fit(X
_tr_p_scaled, y_tr_p)
rf_p = RandomForestClassifier(n_estimators=150, max_depth=5, class_weight='balanced', ra
ndom_state=42).fit(X_tr_p_scaled, y_tr_p)
print(f"Pressure Ulcer ROC-AUC: LR={roc_auc_score(y_te_p, lr_p.predict_proba(X_te_p_scal
ed)[:,1]):.3f}, RF={roc_auc_score(y_te_p, rf_p.predict_proba(X_te_p_scaled)[:,1]):.3f}")
```

Appendix B: Mathematical Formulas for Quality Measures and Logistic Regression

1. CMS Major Injury Fall Rate (KPI_1):

```
Rate_Fall_Injury = ( ∑ I(ANYFALLS = 1 ∧ (HIPFRACT = 1 ∨ OTHFRACT = 1)) /
Total_LongStay_Census ) × 100
```

2. High-Risk Pressure Ulcer Rate (KPI_2):

```
Rate_HighRisk_PU = ( ∑ I(ULCERHI ∈ {2, 3, 4}) / Count(High_Risk_Cohort) )
× 100
```

3. Multivariate Logistic Regression Probability Function:

```
P(Y_i = 1 | x_i) = 1 / ( 1 + exp( - ( β_0 + ∑ β_j × x_ij ) ) )
Adjusted Odds Ratio (OR_j) = exp(β_j) = e^(β_j)
```

Appendix C: Comprehensive Variable Mapping Table

Table C1
Variable Mapping and Data Dictionary Definitions

Variable	Domain	Permissible Values	Cleaned Definition / Transformation
`ANYFALLS`	Clinical	1=Yes, 2=No, 8=Unknown	Binary Target: 1 (Fall Occurred) vs. 0 (No Falls).
`ULCERHI`	Clinical	0=None, 1-4=Stage, 8=Unk	Binary Target: 1 (Stage 1-4 Ulcer) vs. 0 (None).

Variable	Domain	Permissible Values	Cleaned Definition / Transformation
`TOTALADL`	Functional	0 to 5, 8=Unknown	Composite score (0=Independent to 5=Dependent).
`BEDMOBIL`	Functional	0 to 4, 88=Unknown	Bed mobility rating (0=Indep to 4=Total dependence).
`TRANSFER`	Functional	0 to 4, 88=Unknown	Transfer capability (0=Indep to 4=Total dependence).
`SIDERAIL`	Safety Device	0=None, 1=Partial, 2=Full	Binary indicator of bed siderail restraint usage.
`BLADCONT`	Physiological	0 to 4, 8=Unknown	Bladder continence (0=Continent to 4=Incontinent).
`WGTLOSS`	Physiological	1=Yes, 2=No, 8=Unknown	Unintended weight loss (≥5% in 30d / ≥10% in 180d).

Appendix D: Parameter Estimates and Classification Metrics

Table D1

Multivariate Logistic Regression Summary for Fall and Pressure Ulcer Models

Predictor	Fall Model Coef (β)	Fall Odds Ratio (OR)	PU Model Coef (β)	PU Odds Ratio (OR)
`TOTALADL`	0.2160 ($p = 0.007$)	**1.241**	0.0939 ($p = 0.505$)	1.098
`TRANSFER`	0.0074 ($p = 0.951$)	1.007	0.4374 ($p = 0.020$)	**1.549**
`WALKING`	-0.3889 ($p = 0.017$)	0.678	0.5746 ($p = 0.020$)	**1.776**
`BEDMOBIL`	-0.0654 ($p = 0.467$)	0.937	0.2293 ($p = 0.103$)	1.258
`BLADCONT`	0.1414 ($p = 0.027$)	**1.152**	-0.2985 ($p = 0.001$)	0.742
`SIDERAIL`	0.4079 ($p = 0.006$)	**1.504**	-0.0537 ($p = 0.804$)	0.948
`WGTLOSS`	0.5263 ($p = 0.015$)	**1.693**	0.0615 ($p = 0.844$)	1.063
`MOOD`	0.2922 ($p = 0.002$)	**1.339**	-0.0299 ($p = 0.842$)	0.971